APPENDIX II

THE DEMOCRATIC REPUBLIC OF THE CONGO

DEBT SUSTAINABILITY ANALYSIS USING THE LOW-INCOME COUNTRY FRAMEWORK

1. As in the case of the most recent low-income country debt sustainability analysis (LIC DSA) conducted in December 2009, the DRC would continue to face a high risk of debt distress after receiving debt relief from all creditors under the enhanced HIPC Initiative and MDRI. The PV of external debt-to-GDP indicator breaches the policy-dependent threshold in the baseline scenario and the external debt outlook is vulnerable to adverse shocks. However, the breach is relatively small and short-lived and is mainly due to the inclusion of debt related to a public guarantee on external borrowing to finance public infrastructure projects under the SCCA. Without the guarantee, all external debt indicators would remain well below their respective thresholds and the risks to debt sustainability therefore appear to be manageable.

I. BACKGROUND

2. **This LIC DSA for the DRC assesses its public and external debt dynamics using the forward-looking debt sustainability framework (DSF) for low-income countries**. The LIC DSA uses the reconciled debt database prepared for the completion point HIPC DSA, and incorporates the impact of HIPC, additional bilateral assistance beyond HIPC, and MDRI relief in the baseline scenario. The LIC DSA and the HIPC DSA share the same macroeconomic assumptions in the baseline and alternative scenarios but differ in four key areas: (i) the discount rate for the LIC DSA is fixed at 4 percent, compared to the currency-specific six-month averages of commercial interest reference rates for the HIPC DSA; (ii) the LIC DSA uses exchange rate projections from the *World Economic Outlook* instead of actual exchange rates at end-2009 used for the HIPC DSA; (iii) exports used for the debt burden indicators in the LIC DSA are the latest projections rather than three-year backward-looking averages and (iv) the baseline in the LIC DSA assumes that IDA assistance is delivered in the form of concessional loans rather than mainly grants as in the HIPC DSA.

In the LIC DSA framework the PV of the DRC's public and publicly guaranteed (PPG) external debt is US$2.4 billion (23.8 percent of GDP) at end 2009, assuming full delivery of HIPC and MDRI assistance. Around three-fifths of this debt would be owed to multilateral creditors, with the balance owed to official bilateral and commercial creditors almost equally (18 and 20 percent respectively). The bulk of external debt owed to commercial creditors is a public guarantee on loans for public infrastructure projects provided under the terms of the SCCA.[1]

[1] Details of this agreement are outlined in IMF Country Report No. 10/88 (Box 3) and IDA/SecM2010-0285 (Box 2).

3. **The DRC has made good-faith efforts to normalize relations with all external creditors**. The DRC has not accumulated arrears with multilateral creditors since the commencement of the current ECF arrangement in July 2009.[2] In February 2010, Paris Club agreed to reschedule the DRC's debt under exceptional terms, which reduced debt service (including arrears) due by some 97 percent over the period July 2009 to June 2012 (the period covered by the current ECF arrangement). The DRC authorities then requested comparable terms from its other creditors. They have maintained contact with their creditors to keep them informed about ongoing developments with the Paris Club and the HIPC completion point process more generally, and have engaged in negotiations to make interim debt service payments taking into account their limited fiscal capacity.

II. External Debt Sustainability Analysis

4. **The baseline scenario in this DSA assumes a strong recovery in economic activity over the medium term, supported by large investments in mining and public infrastructure projects**. In particular, it assumes that the security situation stabilizes and that the government adopts prudent macroeconomic policies and makes significant progress on key structural reforms, including strengthening tax collection, improving budget preparation and execution, and reforming the civil service. The near-term outlook is dominated by a rebound in world commodity prices (copper in particular) and external demand associated with the global economic recovery, followed by a strong recovery over the medium term fueled by a US$3.2 billion investments in a mining project along with US$3 billion in investments in public infrastructure projects envisaged under the SCCA. Real GDP growth is projected to increase from 2.8 percent in 2009 to an average rate of 6.8 percent in 2010–15 and then gradually decline to 4.4 percent in the long run.

5. **Government revenues (excluding grants) are projected to rebound from 17 percent of GDP in 2009 to 20 percent in 2010, and then increase gradually to just under 24 percent over the long term**. Meanwhile grants drop sharply from 11.7 percent of GDP in 2009 to 6.6 percent in 2010 and then decline gradually to under 4 percent in the long run.

6. **The current account deficit widens dramatically from 10 percent of GDP in 2009 to over 20 percent in 2010–11, mainly due to a large increase in imported factors used in the mining and public infrastructure projects**. The deficit narrows thereafter as mining exports expand and investment-related imports diminish over time, gradually declining below 4 percent of GDP by the end of the 20-year projection horizon. Much of the widening in the current account deficit in 2010–11 is covered by net FDI, which increases from 5.7 percent of GDP in 2000 to 8.6 percent in 2011 and then quickly falls back to 5.6 percent in 2013. The surge in FDI inflows reflects foreign investment in a series of mining projects planned over the next few years (notably those financed under the SCCA).

[2] See discussion of non-accumulation of external arrears in accompanying IMF Staff Report.

7. **External financing in the short term is dominated by loans disbursed to fund public infrastructure projects under the SCCA, which account for over one half of gross borrowing needs over the period 2010–14**. The average grant element of new borrowing rises steadily from 20 percent in 2010–11[3] to just over 48 percent in 2015–16 and then declines gradually to under 45 percent in the long run. This profile reflects three factors: (i) public infrastructure loans disbursed under the SCCA in 2010–14 have a much lower grant element (2 percent) than other sources of external financing; (ii) the IMF purchases under the current ECF arrangement have a lower grant element than funds provided by other multilateral creditors; and (iii) the country makes a gradual transition away depending exclusively on external financing from multilateral creditors to financing from official bilateral creditors as the economy develops over time.

8. **The baseline scenario assumes that the HIPC completion point is reached in June 2010**. Under the assumption that all creditors provide debt relief under the HIPC/MDRI Initiatives, the PV of PPG external debt at end-2009 would be US$2.4 billion, an amount equal to 24.1 percent of GDP, 52.6 percent of exports and 141.1 percent of government revenue (excluding grants). Each of the three external debt indicators is well below the policy-dependent threshold levels of 30 percent, 100 percent and 200 percent, respectively.[4] By contrast, in the most recent LIC DSA,[5] the PV of PPG external debt at end-2009 was US$11.5 billion in the baseline scenario, which assumed that creditors would provide interim HIPC relief but did not assume that the completion point would be reached. In that scenario, all three external debt indicators were well above the thresholds in 2009 (104 percent of GDP, 256 percent of exports and 572 percent of government revenue), and for most of the 20-year projection period, indicating that the DRC was in debt distress.

9. **Under the baseline scenario, one external debt burden indicator breaches the policy-dependent threshold** (see Figure 1). The PV of external debt burden is projected to increase sharply over the medium term, with the PV of external debt reaching US$6.7 billion in 2016, almost three times the level in 2009. The external debt-to-GDP indicator breaches the 30 percent threshold over a six-year period (2011–16), averaging 36 percent in 2012–15.

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[3] The average grant element on new borrowing is well below the 35 percent grant element required to qualify as concessional under the current ECF arrangement. This is because the concessionality calculation used to assess the average grant element under the terms of an IMF program uses a much higher discount rate than that used in the LIC DSA. For example, the public infrastructure financed under the terms of the SCCA have a grant element of 38 percent based on the appropriate discount factor of 6.4 percent for the U.S. dollar, compared to a grant element of only 2 percent based on a discount factor of 4 percent used in the LIC DSA.

[4] With an average CPIA rating of 2.80 in 2006–08, the DRC is classified as having a weak policy framework.

[5] Done in November 2009 IMF Country Report No. 10/88, and IDA/SecM2010-0285.

10. **The significant increase in the external debt indicators over the medium term is mainly due to a public guarantee on external borrowing to finance public infrastructure projects under the SCCA**. Under an alternative (counterfactual) scenario where the DRC government did not provide the public guarantee, all external debt indicators would be well below their respective thresholds (see Figure 2). The public infrastructure loans are repaid from operating profits on a mining project also financed under the SCCA.[6] The inclusion of the public guarantee in the baseline scenario therefore substantially overstates the risk of debt distress.

11. **The baseline projections are subject to significant downside risks**. Given the DRC's high dependence on mining exports, the external account and overall economic outlook remains highly vulnerable to adverse terms-of-trade shocks. Moreover, poor governance, unresolved security issues, weak implementation capacity, and lower-than-envisaged donor support could also worsen the external debt outlook significantly.

12. **The external debt outlook is vulnerable to large adverse shocks**. All three debt burden indicators exceed the thresholds under the most extreme stress tests. The debt profile is most vulnerable to two shocks in particular: (1) less concessional terms of external financing; and (2) a sharp drop in exports in 2010–11 (see Figure 1). With regard to the latter, the external debt-to-exports indicator breaches the threshold by a wide margin and remains above the threshold throughout most of the 20-year projection period. This is because of the large magnitude of the export shock. Under this stress test, exports decline by some 38 percent in 2011–12, compared to a 19 percent increase in the baseline scenario. Under a less extreme stress test where export growth is zero in 2011–12, the external debt-to-export indicator would not breach the threshold. Under the historical scenario, the external debt burden indicators decline rapidly. This reflects the fact that the country did not have access to external financing during the conflict period, which restricted the noninterest current account deficit.

III. Public Debt Sustainability Analysis

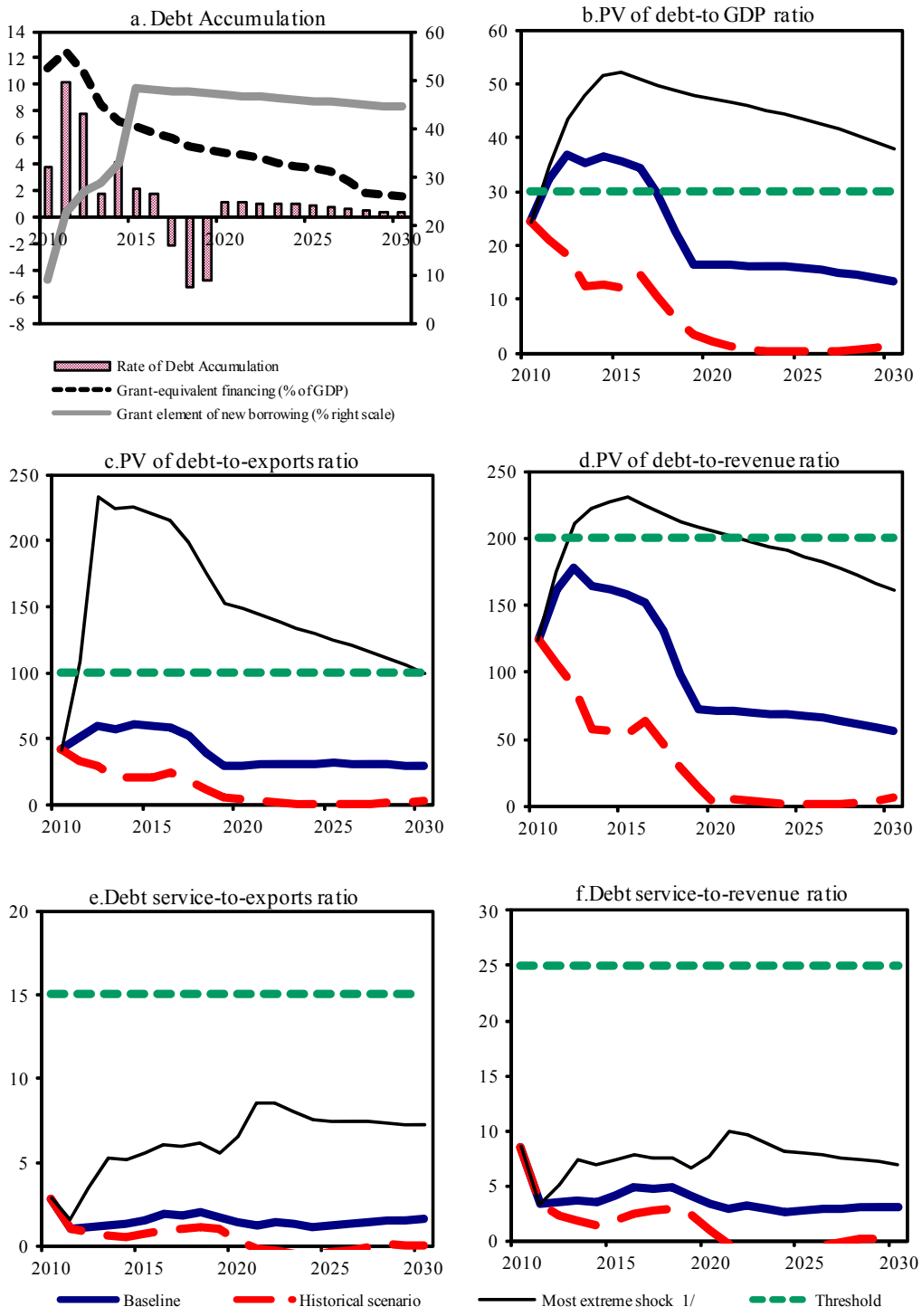
13. **The DRC's domestic debt is relatively low and as a result, the above sustainability analysis for external debt broadly applies to total public debt, albeit with slightly higher debt burden indicators** (see Figure 3). Although information on domestic debt is poor, the authorities recently provided updated estimates on claims of about US$1.2 billion at end-2009 (11 percent of GDP). These claims are mainly from suppliers, public enterprises and public sector employees dating back to the period of conflict in the 1990s. The authorities are in the process of developing a strategy to deal with these claims.

[6] Calculations based on a model of income generated by the mining project developed by the DRC authorities predict that the net operating profits from the mining project would fully repay the public infrastructure loans by 2018, 16 years before the public guarantee would be invoked. Moreover, net operating income would have to decline by 65 percent of the amount projected in the baseline scenario in order for the public guarantee to be invoked.

IV. CONCLUSION

14. **The LIC DSA indicates that the DRC faces a high risk of debt distress after receiving debt relief from all creditors under the enhanced HIPC/MDRI Initiatives**. The external debt-to-GDP indicator breaches the policy-dependent threshold in the baseline scenario. The breach is relatively small and short-lived and is mainly due to a public guarantee on public infrastructure loans under the SCCA, but projections are subject to significant downside risks which tip the balance toward a high risk rating.

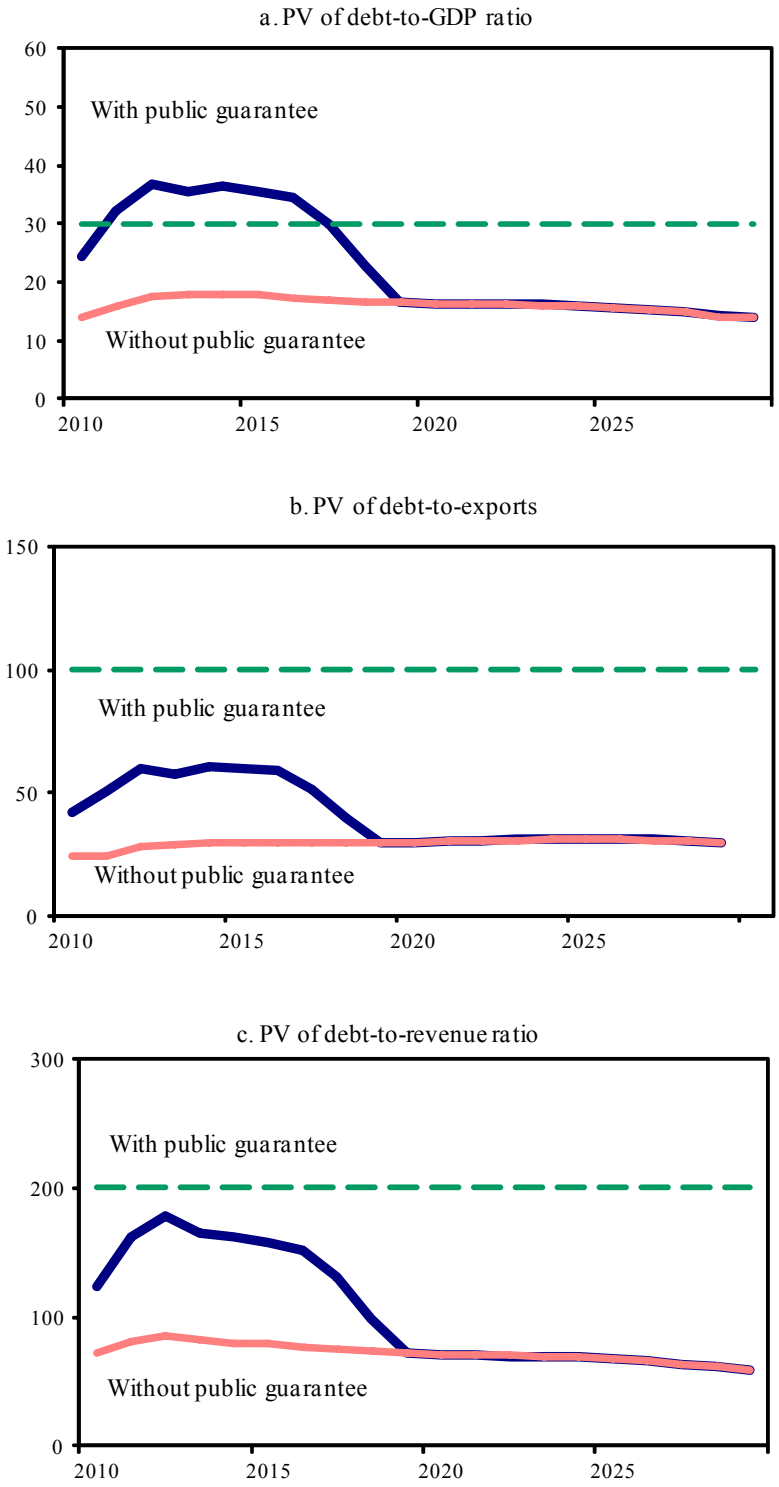
Figure 1. Democratic Republic of the Congo: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-30 1/



Sources: Country authorities; and IMF staff estimates and projections.

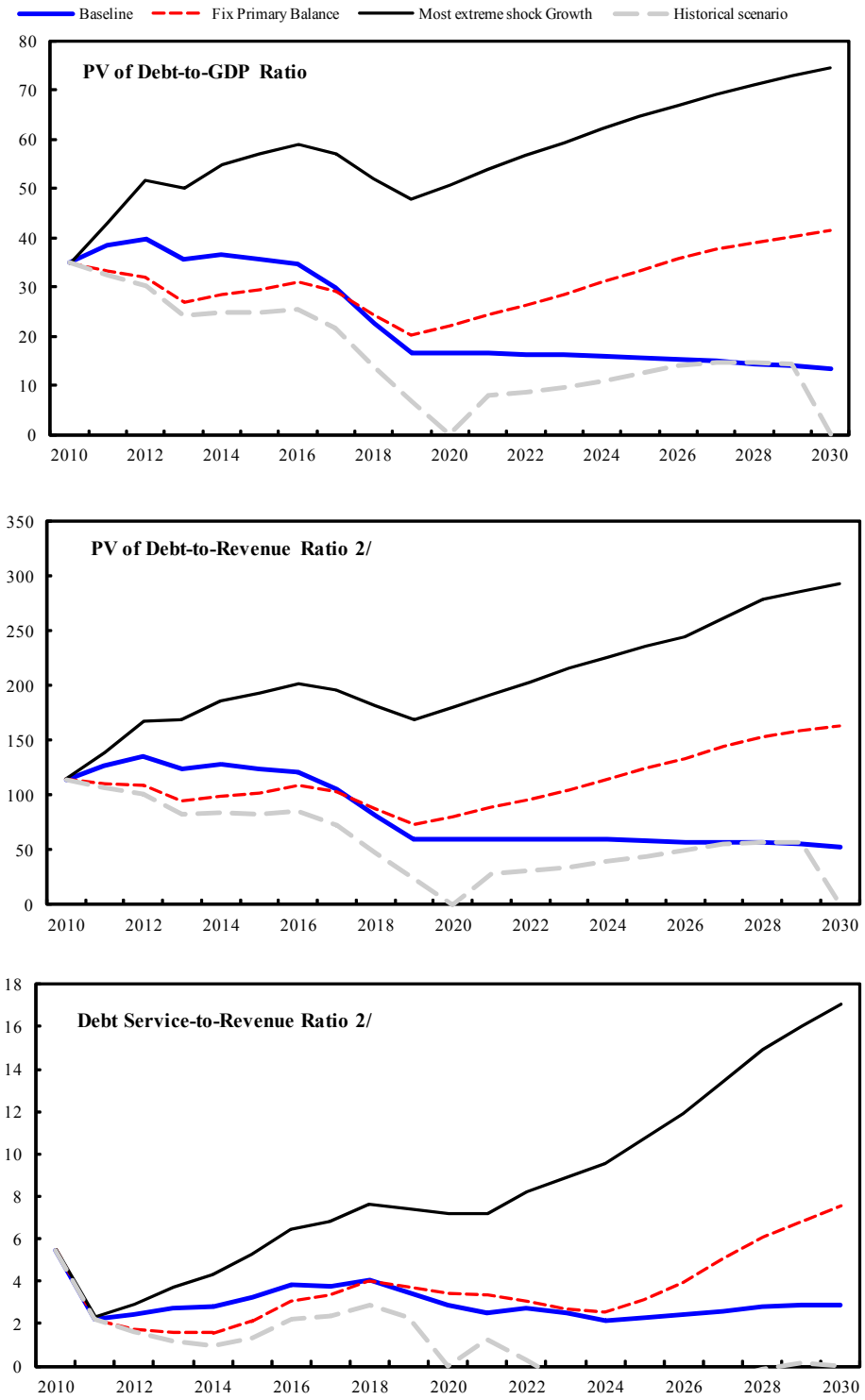
1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Terms shock; in c. to a Exports shock; in d. to a Terms shock; in e. to a Exports shock and in figure f. to a Terms shock

Figure 2. Democratic Republic of the Congo: Indicators of Public and Publicly Guaranteed External Debt with and without Public Guarantee on Public Infrastructure Projects, 2010-30



Sources: Country authorities; and IMF staff estimates and projections.

Figure 3.Democratic Republic of the Congo: Indicators of Public Debt Under Alternative Scenarios, 2010-30 1/



Sources: Country authorities; and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Table 1: Democratic Republic of the Congo: External Debt Sustainability Framework, Baseline Scenario, 2007-30 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**125.7**	**132.5**	**21.1**			**25.5**	**35.0**	**41.1**	**43.6**	**45.2**	**44.7**		**26.5**	**20.6**	
o/w public and publicly guaranteed (PPG)	125.7	132.5	21.1			25.5	35.0	41.1	43.6	45.2	44.7		26.5	20.6	
Change in external debt	-22.9	6.8	-111.3			4.4	9.5	6.1	2.5	1.6	-0.5		-0.2	-1.0	
Identified net debt-creating flows	-20.2	-16.0	12.6			13.4	11.3	8.4	5.4	0.2	-1.0		1.3	-1.4	
Non-interest current account deficit	**-2.5**	**12.7**	**7.1**	**4.7**	**4.6**	**20.4**	**21.3**	**17.8**	**13.2**	**7.3**	**6.3**		**7.2**	**3.8**	4.7
Deficit in balance of goods and services	3.4	15.1	15.9			19.7	20.0	15.8	15.5	9.5	8.4		9.0	4.0	
Exports	65.2	61.3	45.8			58.2	63.1	61.6	61.3	60.4	59.5		54.8	45.5	
Imports	68.6	76.4	61.7			77.9	83.0	77.4	76.8	69.9	67.8		63.9	49.5	
Net current transfers (negative = inflow)	-8.6	-10.6	-12.8	-7.4	2.8	-7.3	-6.6	-6.3	-6.1	-6.0	-5.9		-5.4	-4.3	-5.1
o/w official	-7.4	-8.8	-11.7			-6.6	-5.9	-5.5	-5.4	-5.3	-5.2		-4.7	-3.6	
Other current account flows (negative = net inflow)	2.7	8.2	3.9			8.0	8.0	8.2	3.9	3.8	3.8		3.6	4.2	
Net FDI (negative = inflow)	**-3.7**	**-14.8**	**-5.7**	**-5.0**	**4.0**	**-6.6**	**-8.6**	**-7.4**	**-5.6**	**-5.4**	**-5.3**		**-5.0**	**-4.6**	-4.8
Endogenous debt dynamics 2/	**-13.9**	**-13.9**	**11.3**			**-0.4**	**-1.5**	**-2.0**	**-2.3**	**-1.8**	**-2.0**		**-0.9**	**-0.7**	
Contribution from nominal interest rate	3.9	3.1	3.6			0.6	0.2	0.2	0.8	0.9	0.9		0.2	0.2	
Contribution from real GDP growth	-8.2	-6.7	-4.0			-1.0	-1.6	-2.2	-3.0	-2.7	-2.9		-1.2	-0.9	
Contribution from price and exchange rate changes	-9.7	-10.3	11.7			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-2.7**	**22.8**	**-124.0**			**-9.0**	**-1.8**	**-2.3**	**-2.9**	**1.4**	**0.5**		**-1.5**	**0.4**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	24.1			24.4	32.2	36.8	35.4	36.6	35.6		16.4	13.3	
In percent of exports	...	...	52.6			42.0	51.0	59.7	57.8	60.6	59.9		29.9	29.1	
PV of PPG external debt	**...**	**...**	**24.1**			**24.4**	**32.2**	**36.8**	**35.4**	**36.6**	**35.6**		**16.4**	**13.3**	
In percent of exports	**...**	**...**	**52.6**			**42.0**	**51.0**	**59.7**	**57.8**	**60.6**	**59.9**		**29.9**	**29.1**	
In percent of government revenues	**...**	**...**	**141.1**			**124.3**	**161.7**	**178.1**	**164.6**	**161.9**	**157.8**		**71.2**	**56.2**	
Debt service-to-exports ratio (in percent)	**12.9**	**12.0**	**18.5**			**2.9**	**1.0**	**1.2**	**2.4**	**6.8**	**11.3**		**1.4**	**1.6**	
PPG debt service-to-exports ratio (in percent)	**12.9**	**12.0**	**18.5**			**2.9**	**1.0**	**1.2**	**1.3**	**1.3**	**1.6**		**1.4**	**1.6**	
PPG debt service-to-revenue ratio (in percent)	**57.2**	**39.9**	**49.7**			**8.5**	**3.3**	**3.5**	**3.6**	**3.5**	**4.1**		**3.4**	**3.1**	
Total gross financing need (Millions of U.S. dollars)	215.5	609.4	1075.6			1884.7	1766.2	1582.5	1411.2	1021.2	1399.6		812.4	6.8	
Non-interest current account deficit that stabilizes debt ratio	20.4	5.8	118.4			16.0	11.8	11.7	10.7	5.7	6.8		7.4	4.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.3	6.2	2.8	3.5	4.6	5.4	7.0	6.8	8.1	6.7	7.0	6.8	4.7	4.4	4.7
GDP deflator in US dollar terms (change in percent)	7.0	8.9	-8.1	6.4	8.7	5.6	1.1	1.2	0.9	1.6	1.6	2.0	2.6	2.6	2.6
Effective interest rate (percent) 5/	3.0	2.9	2.6	1.7	1.3	3.4	0.8	0.6	2.1	2.3	2.2	1.9	0.9	0.9	0.9
Growth of exports of G&S (US dollar terms, in percent)	138.3	8.7	-29.4	22.9	44.5	41.4	17.3	5.5	8.5	6.7	7.0	14.4	5.7	5.0	5.5
Growth of imports of G&S (US dollar terms, in percent)	83.7	28.8	-23.6	27.2	27.3	40.4	15.3	0.7	8.1	-1.4	5.4	11.4	15.2	4.4	5.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	8.8	22.8	27.2	28.9	33.0	48.4	28.2	47.1	44.5	46.3
Government revenues (excluding grants, in percent of GDP)	14.7	18.5	17.1			19.6	19.9	20.7	21.5	22.6	22.6		23.1	23.6	23.2
Aid flows (in Millions of US dollars) 7/	429.4	359.5	874.4			1394.3	1891.0	1804.7	1587.9	1494.8	1540.7		1600.1	1113.2	
o/w Grants	225.6	267.5	764.6			1294.3	1303.6	1206.2	1049.6	951.9	974.8		1008.8	617.0	
o/w Concessional loans	203.8	92.0	109.8			100.0	587.4	598.5	538.4	542.9	565.9		591.3	496.2	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			11.3	12.5	10.9	8.5	7.3	6.8		4.9	1.6	4.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			61.4	58.8	63.0	66.5	69.0	81.0		80.4	75.2	79.4
Memorandum items:															
Nominal GDP (Millions of US dollars)	10028	11595	10958			12200	13193	14249	15539	16841	18297		26530	52959	
Nominal dollar GDP growth	13.7	15.6	-5.5			11.3	8.1	8.0	9.1	8.4	8.6	8.9	7.5	7.1	7.3
PV of PPG external debt (in Millions of US dollars)			2388.4			2806.4	4050.7	5070.9	5327.4	5971.6	6333.2		4271.5	6893.4	
(PVt-PVt-1)/GDPt-1 (in percent)						3.8	**10.2**	**7.7**	1.8	4.1	2.1	5.0	1.1	0.3	-0.1
Gross remittances (Millions of US dollars)	…	…	…			…	…	…	…	…	…		…	…	
PV of PPG external debt (in percent of GDP + remittances)	...	...	24.1			24.4	32.2	36.8	35.4	36.6	35.6		16.4	13.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	52.6			42.0	51.0	59.7	57.8	60.6	59.9		29.9	29.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	18.5			2.9	1.0	1.2	1.3	1.3	1.6		1.4	1.6	

Sources: Country authorities; and IMF staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ?(1+g)]/(1+g+?+g?) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ? = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Democratic Republic of the Congo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
PV of debt-to GDP ratio								
Baseline	24	32	37	35	37	36	**16**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	24	21	18	12	13	12	**2**	2
A2. New public sector loans on less favorable terms in 2010-2030 2	24	35	43	48	51	52	**47**	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	24	33	42	40	41	40	**19**	15
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	24	45	71	68	68	65	**41**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	24	32	38	37	38	37	**17**	14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	24	37	48	46	46	45	**24**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	24	39	50	48	49	47	**25**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	24	44	51	49	50	49	**23**	18
PV of debt-to-exports ratio								
Baseline	42	51	60	58	61	60	**30**	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	42	33	30	20	21	20	**4**	3
A2. New public sector loans on less favorable terms in 2010-2030 2	42	55	71	78	85	88	**86**	83
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	42	49	58	56	59	58	**29**	28
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	42	108	233	224	226	221	**149**	100
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	42	49	58	56	59	58	**29**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	42	59	77	74	77	75	**44**	35
B5. Combination of B1-B4 using one-half standard deviation shocks	42	62	73	71	73	72	**41**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	42	49	58	56	59	58	**29**	28
PV of debt-to-revenue ratio								
Baseline	124	162	178	165	162	158	**71**	56
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	124	106	89	58	57	53	**0**	6
A2. New public sector loans on less favorable terms in 2010-2030 2	124	175	211	222	228	231	**205**	161
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	124	167	201	186	183	179	**81**	64
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	124	228	346	317	300	289	**176**	96
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	124	160	185	170	168	164	**74**	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	124	188	230	212	205	199	**105**	68
B5. Combination of B1-B4 using one-half standard deviation shocks	124	197	241	222	215	209	**108**	72
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	124	220	245	226	223	218	**99**	78
Debt service-to-exports ratio								
Baseline	3	1	1	1	1	2	**1**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	3	1	1	1	0	1	**0**	0
A2. New public sector loans on less favorable terms in 2010-2030 2	3	1	2	6	6	7	**5**	5
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	3	1	1	1	1	2	**1**	2
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	3	2	3	5	5	6	**7**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	3	1	1	1	1	2	**1**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	3	1	1	2	2	2	**2**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	3	1	1	2	2	2	**2**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	3	1	1	1	1	2	**1**	2
Debt service-to-revenue ratio								
Baseline	8	3	3	4	4	4	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	8	3	2	2	1	2	**1**	0
A2. New public sector loans on less favorable terms in 2010-2030 2	8	3	7	17	17	17	**12**	9
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	8	4	4	4	4	5	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	8	3	5	7	7	7	**8**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	8	3	4	4	4	4	**4**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	8	3	4	5	5	5	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	8	3	5	5	5	5	**5**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	8	5	5	5	5	6	**5**	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	30	30	30	30	30	30	**30**	30

Sources: Country authorities; and IMF staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Democratic Republic of the Congo: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-30
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	125.9	145.0	34.3			36.0	41.1	43.9	43.6	45.2	44.7		26.5	20.6	
o/w foreign-currency denominated	125.7	132.5	21.1			25.5	35.0	41.1	43.6	45.2	44.7		26.5	20.6	
Change in public sector debt	-23.3	19.1	-110.7			1.6	5.2	2.8	-0.3	1.6	-0.5		-0.2	-1.0	
Identified debt-creating flows	-31.5	0.8	-1.0			-3.7	4.6	2.4	0.8	0.0	-1.7		-0.5	-0.5	
Primary deficit	-1.7	-1.0	-0.2	1.6	2.9	3.3	8.5	6.1	4.4	3.3	1.7	4.5	1.2	0.7	0.7
Revenue and grants	17.0	20.8	24.0			30.4	30.1	29.4	28.5	28.5	28.7		27.5	25.3	
of which: grants	2.2	2.3	7.0			10.6	9.9	8.5	6.8	5.7	5.3		3.8	1.2	
Primary (noninterest) expenditure	15.2	19.8	23.9			33.7	38.6	35.5	33.0	31.8	30.4		28.7	26.0	
Automatic debt dynamics	-26.6	4.5	4.1			-5.4	-3.8	-3.7	-3.6	-3.2	-3.5		-1.6	-1.3	
Contribution from interest rate/growth differential	-14.1	-13.9	7.9			-4.6	-3.5	-3.3	-3.6	-3.2	-3.4		-1.6	-1.3	
of which: contribution from average real interest rate	-5.3	-6.6	11.9			-2.9	-1.2	-0.7	-0.3	-0.4	-0.4		-0.4	-0.3	
of which: contribution from real GDP growth	-8.8	-7.3	-4.0			-1.8	-2.3	-2.6	-3.3	-2.7	-3.0		-1.2	-0.9	
Contribution from real exchange rate depreciation	-12.6	18.4	-3.8			-0.8	-0.3	-0.4	0.0	-0.1	-0.1		...	...	
Other identified debt-creating flows	-3.1	-2.7	-4.9			-1.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-3.1	-2.7	-4.9			-1.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.2	18.3	-109.7			5.3	0.5	0.4	-1.2	1.6	1.2		0.3	-0.5	
Other Sustainability Indicators															
PV of public sector debt	0.2	12.5	37.3			34.9	38.3	39.6	35.4	36.6	35.6		16.4	13.3	
o/w foreign-currency denominated	0.0	0.0	24.1			24.4	32.2	36.8	35.4	36.6	35.6		16.4	13.3	
o/w external	...	...	24.1			24.4	32.2	36.8	35.4	36.6	35.6		16.4	13.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.7	6.4	8.3			5.0	9.1	6.8	5.2	4.1	2.7		1.9	1.5	
PV of public sector debt-to-revenue and grants ratio (in percent)	1.4	60.4	155.0			114.5	127.1	134.7	124.2	128.3	124.2		59.7	52.5	
PV of public sector debt-to-revenue ratio (in percent)	1.6	68.0	218.3			175.7	189.1	189.0	162.7	160.1	152.5		69.2	55.1	
o/w external 3/	…	…	141.1			123.0	158.8	175.5	162.7	160.1	152.5		69.2	55.1	
Debt service-to-revenue and grants ratio (in percent) 4/	49.6	35.5	35.3			5.5	2.2	2.4	2.7	2.8	3.2		2.8	2.9	
Debt service-to-revenue ratio (in percent) 4/	57.2	39.9	49.7			8.4	3.3	3.4	3.6	3.5	4.0		3.3	3.0	
Primary deficit that stabilizes the debt-to-GDP ratio	21.6	-20.1	110.5			1.7	3.3	3.3	4.8	1.7	2.3		1.3	1.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.3	6.2	2.8	3.5	4.6	5.4	7.0	6.8	8.1	6.7	7.0	6.8	4.7	4.4	4.7
Average nominal interest rate on forex debt (in percent)	3.0	2.9	2.6	1.7	1.3	3.4	0.8	0.6	0.6	0.5	0.5	1.1	0.9	0.9	0.8
Average real interest rate on domestic debt (in percent)	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Real exchange rate depreciation (in percent, + indicates depreciatior	-9.3	16.5	-2.6	-4.1	19.9	-4.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	17.9	18.9	33.3	105.7	184.1	24.3	12.6	9.7	8.0	8.5	7.9	11.8	6.7	6.7	6.9
Growth of real primary spending (deflated by GDP deflator, in perce	-0.1	0.4	0.2	0.2	0.2	0.5	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	8.8	22.8	27.2	28.9	33.0	48.4	28.2	47.1	44.5	...

Sources: Country authorities; and IMF staff estimates and projections.
1/ Central government gross debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Democratic Republic of the Congo: Sensitivity Analysis for Key Indicators of Public Debt 2010-30

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	35	38	40	35	37	36	16	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	35	32	30	24	25	25	0	0
A2. Primary balance is unchanged from 2010	35	33	32	27	28	29	22	41
A3. Permanently lower GDP growth 1/	35	39	41	38	40	40	30	62
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	35	43	51	50	55	57	51	75
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	35	34	34	30	32	31	12	10
B3. Combination of B1-B2 using one half standard deviation shocks	35	35	35	33	37	39	30	48
B4. One-time 30 percent real depreciation in 2011	35	49	49	42	43	42	22	19
B5. 10 percent of GDP increase in other debt-creating flows in 2011	35	48	49	45	45	44	24	19
PV of Debt-to-Revenue Ratio 2/								
Baseline	115	127	135	124	128	124	60	53
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	115	107	101	83	84	83	0	0
A2. Primary balance is unchanged from 2010	115	110	108	94	100	102	80	164
A3. Permanently lower GDP growth 1/	115	129	138	131	139	140	107	242
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	115	139	167	169	186	193	180	293
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	115	114	116	106	111	108	45	41
B3. Combination of B1-B2 using one half standard deviation shocks	115	113	114	112	127	132	108	189
B4. One-time 30 percent real depreciation in 2011	115	161	165	148	151	146	78	74
B5. 10 percent of GDP increase in other debt-creating flows in 2011	115	160	167	156	159	154	85	74
Debt Service-to-Revenue Ratio 2/								
Baseline	5	2	2	3	3	3	3	3
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	2	2	1	1	1	0	0
A2. Primary balance is unchanged from 2010	5	2	2	2	2	2	3	8
A3. Permanently lower GDP growth 1/	5	2	2	3	3	4	4	12
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	5	2	3	4	4	5	7	17
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	5	2	2	2	2	3	2	2
B3. Combination of B1-B2 using one half standard deviation shocks	5	2	2	2	2	3	4	10
B4. One-time 30 percent real depreciation in 2011	5	3	4	4	4	5	5	6
B5. 10 percent of GDP increase in other debt-creating flows in 2011	5	2	4	4	4	4	4	5

Sources: Country authorities; and IMF staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.